



Before You Act: Benchmarking Interpretability in Vision-Language Models

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INTRODUCTION

Large vision-language models (VLMs) have achieved strong multimodal reasoning performance, yet their internal decision-making processes remain unclear. Mechanistic Interpretability (MI) aims to uncover the circuits and representations responsible for model behavior. Acting on these internals, whether to steer, edit, or audit a model, first requires knowing which MI methods recover genuine causal mechanisms rather than surface correlations, yet no such validation exists for multimodal systems. While there are such benchmarks for large language models (LLMs), such as mechanistic interpretability benchmark (MIB) [1] that allow for a rigorous evaluation of MI methods, no similar standardized framework exist for multimodal architectures, so interpretability studies in VLMs remain difficult to compare and validate.

In this work, we introduce Multimodal Mechanistic Interpretability Benchmark (MMIB), a benchmark for evaluating MI methods in VLMs, extending the dual track framework of MIB to multimodal settings. We evaluate two models using two tracks, and add a metric that computes the cross modal information flow. The first track evaluates circuit localization, the second track evaluates causal variable localization. We introduce image- and text-based counterfactual inputs, enabling structured causal validation across both modalities. to ground in its visual input.

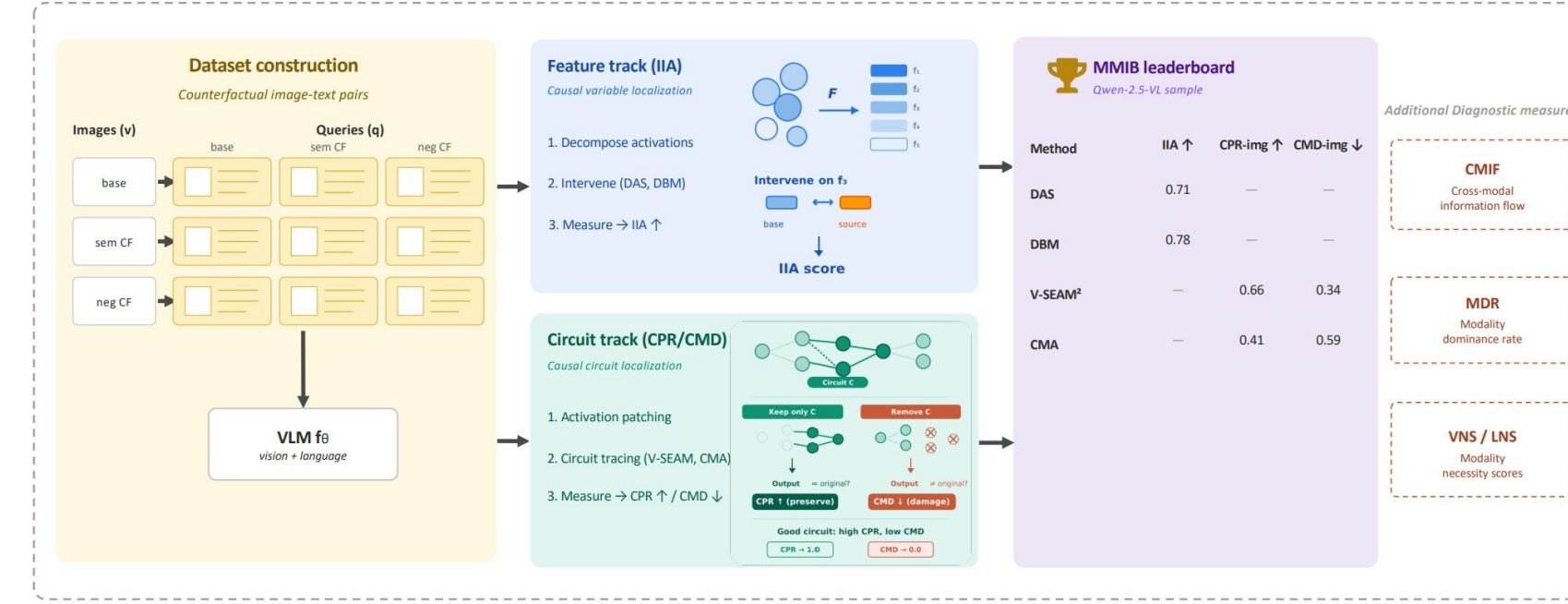


Figure 1. An overview of the (MMIB) workflow.

METHODS

We test two models representing two different architectures:

- Cross-Attention Fusion (e.g., BLIP VQA [2]): where visual features are integrated into the language stream through cross-attention layers.
- Projection-based: where a visual encoder and a projection layer map visual tokens into the embedding space of an autoregressive language model, such as Qwen 2.5-VL-7B-Instruct [3]

We evaluate the models using a visual question answering (VQA), CLEVR-style [4] task. We develop a counterfactual input whose text or image differs from the original one only in the semantic concept.

We evaluate the models on two tracks.

- **Circuit localization:** This track studies the problem of identifying circuits responsible for task behavior in vision-language models.
- **Causal Variable Localization:** This track studies identification of internal representations that causally implement a target concept in vision-language models.

We also extend MIB [1], by adding several multimodality-specific metrics.

1. **Cross-Modal Information Flow (CMIF):** Measures the proportion of the causal decision-making within a model that occurs in the cross-modal bridge.
2. **Modality Dominance Rate and Decision Consistency (MDR & DC):** MDR Quantifies how models resolve conflicting multimodal inputs by measuring the frequency with which the prediction aligns with either a visual cue or a textual cue (or neither), **DC:** quantifies how frequently the model chooses one of the two modalities rather than producing an unrelated answer.
3. **Modality Necessity Scores (VNS, LNS):** Measure how essential a modality is to a prediction by calculating the fraction of changed answers when the modality input is ablated.

RESULTS

For the Circuit Localization track we use **CPR** (integrated circuit performance ratio) and **CMD** (integrated circuit-model distance) metrics. To account two streams in multimodal architectures, we report text-altered and image-altered CPR/CMD. We use the metrics to evaluate the following methods: **CMA**, Visual Semantic Editing and Attention Modulating (**V-SEAM**) and a **random component selection** baseline.

For the Causal Variable Localization track, we use the **IIA** (Interchange Intervention Accuracy, measures “intervention success” on target concepts.)

We test the following interchange strategies: **Full Vector Swap**, **PCA interventions**, **Differential Binary Masking (DBMN)**, and **Distributed Alignment Search (DAS)**.

1. **In Circuit Localization** track, V-SEAM successfully localizes both image- and text-dependent circuits with high faithfulness. Random method displays unexpectedly high performance, suggesting redundancy in large vision language models, and distributed nature of computation in multimodal representations.
2. **In Causal Variable Localization** track Layer-wise IIA heatmaps reveal that causal variables are most faithfully implemented in vision encoder layers, which is expected since we are using image counterfactuals with faithfulness gradually increasing in deeper multimodal and language components, indicating progressive integration of visual information.
3. **Modality-specific metrics** show that the architecture of the model heavily influences how it integrates modalities and how it handles modality conflict. 72-82% of causal importance in BLIP VQA [2] is routed through its cross-attention bridge, where cross-modal flow strongly predicts trust in visual cues under conflict. The decoder-only model LLaVA integrates visual information in the deeper layers visual input is necessary for non-conflicting predictions (high VNS) yet is consistently overridden by the textual priors in the presence of modality conflict.

(a) Modality dominance			
Model	MDR _v	MDR _t	DC
BLIP	0.312	0.210	0.598
LLaVA	0.078	0.538	0.873
Qwen	0.338	0.274	0.552

(b) Causal localization			
Method	BLIP	LLaVA	Qwen
Normal	0.261	0.148	0.984
PCA	0.232	0.136	0.009
DAS	0.235	0.071	0.709
DBM	0.208	0.122	0.782
PCA+DBM	0.203	0.126	0.003

Method	CPR-I	CMD-I	CPR-T	CMD-T
BLIP-Encoder				
Random	0.786	0.225	0.878	0.141
CMA	0.888	0.113	0.931	0.070
VSEAM ¹	0.411	0.594	0.600	0.448
VSEAM ²	0.969	0.045	0.935	0.099
BLIP-Decoder				
Random	0.627	0.374	0.590	0.410
CMA	0.985	0.015	0.988	0.012
VSEAM ¹	0.096	0.904	0.092	0.908
VSEAM ²	0.972	0.028	0.745	0.255
Qwen-2.5-VL				
Random	0.826	0.174	0.966	0.034
CMA	0.405	0.595	0.995	0.006
VSEAM ¹	0.838	0.164	0.927	0.076
VSEAM ²	0.659	0.341	0.954	0.048

¹ Attention-head interventions. ² Activation patching. I/T denote image/text interventions.

Table 1: Modality dominance, causal localization, and CPR/CMD results.

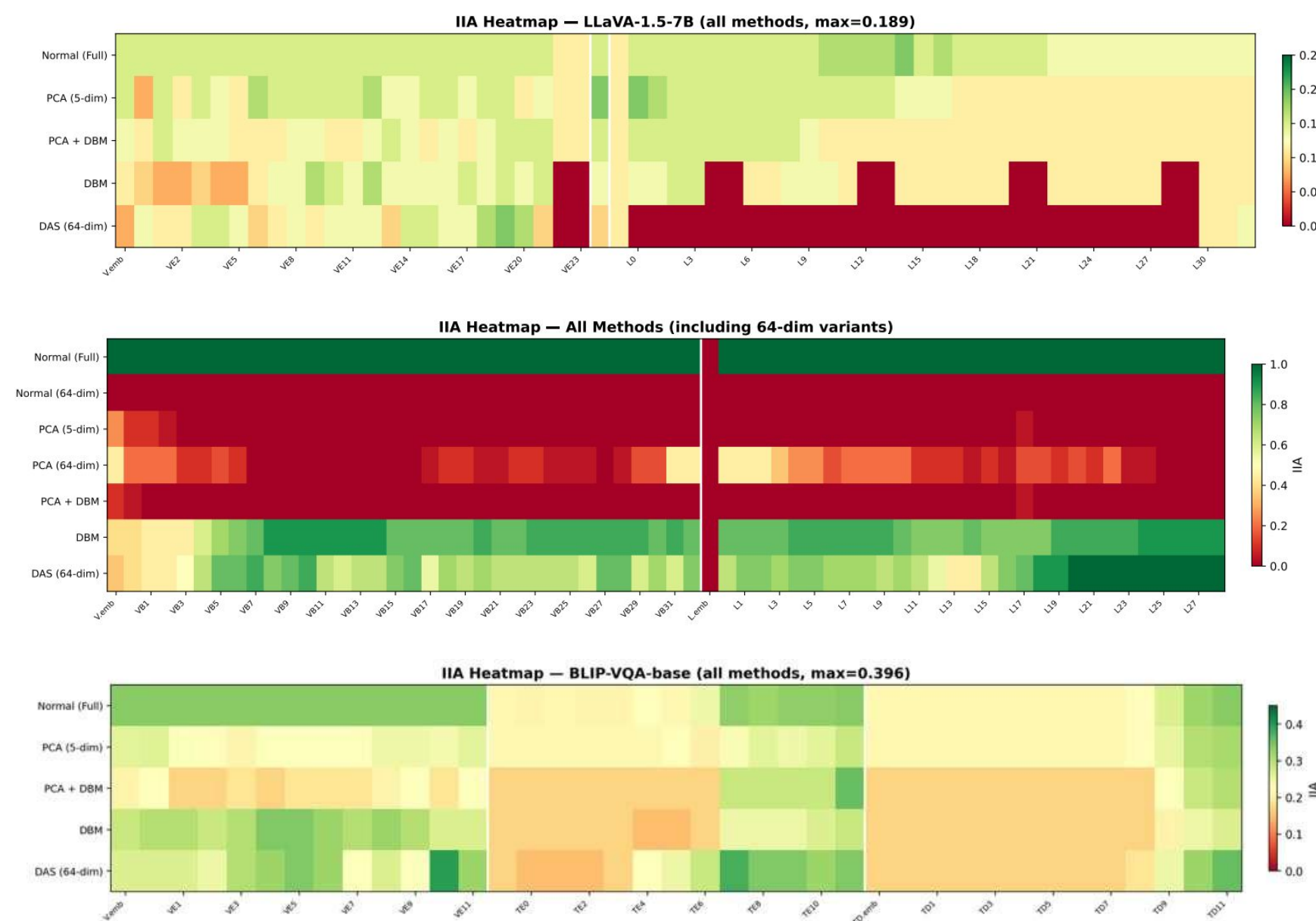


Figure 2: Modality dominance, causal localization, and CPR/CMD results. IIA heatmaps. Layer-wise Interchange Intervention Accuracy across BLIP, LLaVA, and Qwen.

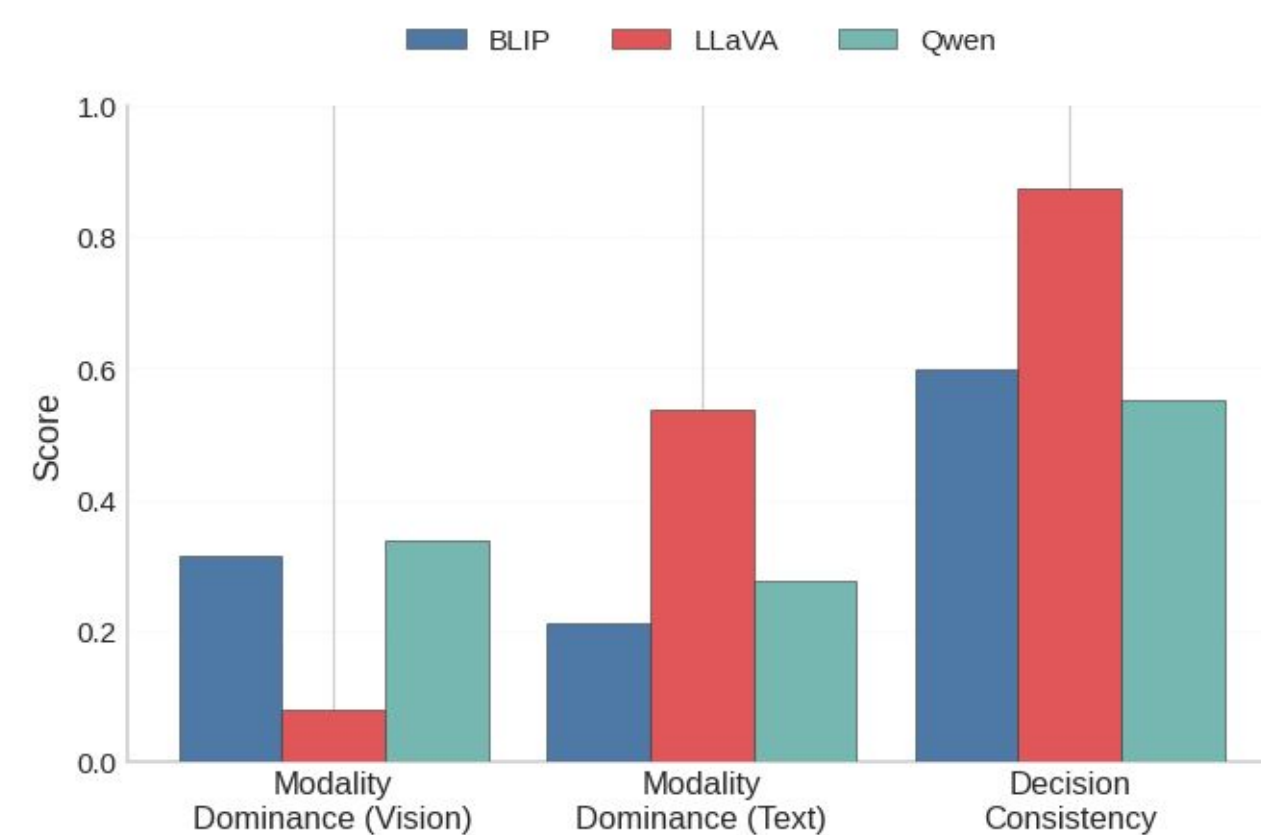


Figure 3: Modality dominance and decision consistency.

DATASET

To enable controlled evaluation of multimodal reasoning, we construct a **synthetic counterfactual dataset** using a procedural rendering pipeline built on Blender assets derived from CLEVR [4].

Synthetic generation allows for precise manipulation of the scene, and more importantly, of the target semantic concept, while also eliminating real-world spurious correlations present in the image. The objects are randomly distributed across the scene (within certain constraints like distance or absence of overlap). Then, using the metadata of the scene, counterfactual images are generated.

- **Semantic Counterfactuals (Causal Reasoning)** - Counterfactuals where the semantic concept of the target object is changed (thus, changing the ground truth)
- **Negative Counterfactuals** - Preserves the semantics, altering the image distribution (e.g. adding noise or render distortions)

Dataset Statistic	Value
Total Semantic CF Pairs	200 (10 types $\times$ 20)
Negative CF Pairs	200 (8 types $\times$ 25)
Image Resolution	720 $\times$ 1280
Generation Method	Procedural + rejection sampling

Table 2: Summary of the MMIB evaluation dataset.

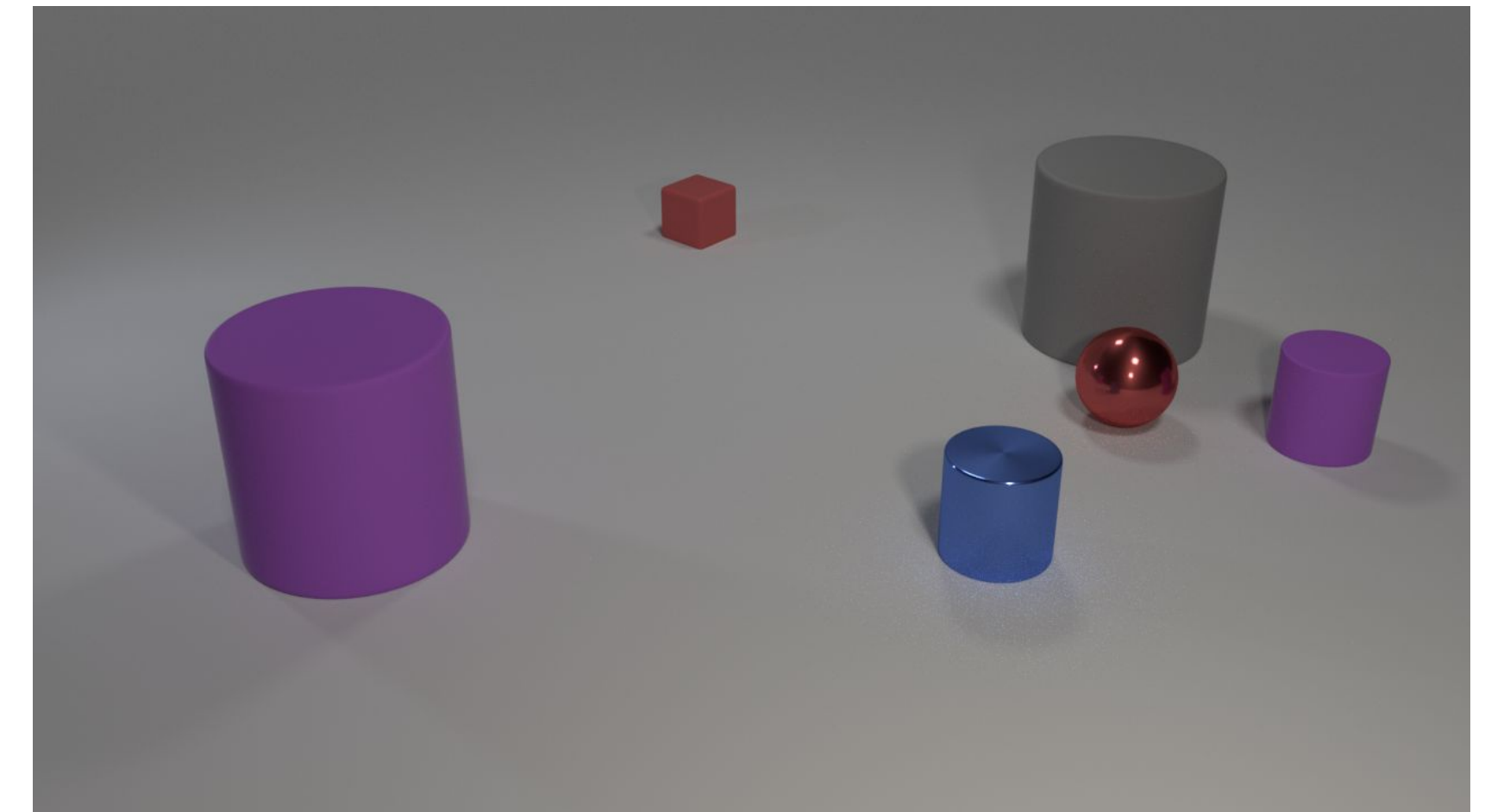


Figure 4: Example image from the dataset.

CONCLUSION

We introduced MMIB, a benchmark for evaluating mechanistic interpretability methods in vision-language models. By extending circuit localization and causal variable localization to modality-specific counterfactual settings, we establish a structured framework for multimodal causal validation.

Our work demonstrates that tracks and faithfulness metric remain relevant in multimodal architecture, and modality-specific metrics can provide a necessary deployment relevant insight into the behaviour of multimodal architectures. MMIB will continue to be a live benchmark as mechanistic interpretability advances and we hope can serve as a foundation for future state-of-the-art methodologies.

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